From dense formula to real spike traces

TODAY: A DENSE FORMULA

- cycle count = product of loop factors
- zero notion of real spike sparsity
- same tile size -> same cycles, always

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ARCHITECTURES DIFFER

- SpinalFlow, PTB, LoAS, Prosperity,
- Phi, GustavSNN process spikes differently
- dense formula can't capture any of it

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PLUGGABLE ArchComputeModel

- Protocol: format_input + compute_cycles
- NodeTileSpec: which trace slice a tile covers
- old formula = just the default plugin

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FIRST PLUGIN: SPINALFLOW

- reconstructs real spike 'spine', time-sorted
- pulled from a real captured LoAS trace
- cycle count = spine length

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WHERE THIS SITS

- Part 2 of a bigger effort (Part 1: pages 1-3)
- only SpinalFlow built so far
- 5 more archs: future work, same interface

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